

Internship Project Report: Robotic Arm Simulation

Intern: Internship Candidate

Task: 2 – Robotic Arm Simulation

Software Used: RoboDK

1. Project Objective

The objective of this task was to design a motion simulation for a robotic arm to perform a "Pick-and-Place" operation. This task demonstrates the ability to manage robot kinematics, target definition, and simulation events in a professional robotics environment.

2. Robotic System Specifications

Robot Model: Universal Robots UR5

Degrees of Freedom (DoF): The UR5 is a 6-DoF industrial manipulator.

Definition: Degrees of Freedom represent the number of independent axes or joints that allow the robot to move.

Configuration: The UR5 features 6 revolute joints (rotational axes), allowing it to reach any point (X, Y, Z) within its spherical workspace and orient its end-effector in any direction (Roll, Pitch, Yaw).

3. Simulation Workflow

To achieve the simulation, the following steps were performed in RoboDK:

- **Model Import:** Imported the UR5 model from the RoboDK online library.
- **Station Setup:** Defined the robot base as the reference frame and attached a gripper and an O-ring (object) to the robot's flange.
- **Target Definition:** Created three primary targets: Home, Pick, and Place.
- **Program Logic (Prog1):** Implemented motion commands (MoveJ, MoveL) combined with Attach/Detach simulation events.

4. Technical Observations

A hybrid approach was used, utilizing *MoveJ* for high-speed transit between points and *MoveL* for precise linear movement during the picking and placing phases. Kinematic reachability was verified by adjusting joint configurations to avoid mechanical limits.

5. Conclusion

The simulation was successfully completed. The robot correctly follows the program logic, interacts with the object, and returns to the home position, meeting all requirements for an industrial pick-and-place operation.